Mrugank Dake

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WORK EXPERIENCE

Postdoctoral Researcher, Dartmouth College **June 2026 – present**
Psychological and Brain Sciences, Advisor: Dr. John Murray

- Learning & Attention: Study how attention shapes what people learn in novel reinforcement-learning tasks, using kernel methods (neural tangent kernel) to compare human and artificial learners.
- Computational Modeling: Build neural network and dynamical-systems models of neural data to probe credit assignment, generalization and inductive biases.

Product Data Science Intern (PhD), Meta Reality Labs **May 2025 – August 2025**

- Product Analytics: Investigated hand-tracking usage across the Quest ecosystem with Presto and Spark, querying large user datasets to find friction points and drive product insights.
- Machine Learning: Built Python models to isolate demographic and behavioral drivers of user drop-off, leading to targeted UX recommendations for Quest applications.
- Network Analysis: Applied graph-theory methods to measure how social networks influence Mixed Reality (MR) adoption and long-term hand-tracking retention.
- Cross-functional Impact: Worked with XFN partners to refine topline metrics and shape the roadmap for next-generation hardware; tested pre-release VR/MR prototypes.

Graduate Teaching Assistant, New York University **January 2023 – December 2025**
Courses: Scientific Programming (MA), iEEG/EEG/MEG Methods (PhD), Introduction to Cognitive Neuroscience (Undergraduate), Math Tools III: Linear Systems & Fourier Transforms (PhD)

- Led discussions and tutorials for 150+ students, turning linear algebra, signal processing and Fourier analysis into practical intuition.
- Wrote Python and MATLAB assignments and interactive tutorials; published the iEEG/EEG/MEG Methods course materials as open notebooks (github.com/mindemory/EEG_MEG_Methods).
- Built custom animations to make abstract linear-systems concepts visual, and ran grading for technical assignments and research papers.

EDUCATION

New York University New York, NY
PhD in Computational Cognitive Neuroscience **May 2026**
MPhil in Experimental Psychology **May 2025**
GPA: 3.95/4.0
Relevant Coursework: Math Tools for Cognitive & Neural Science, Psychophysics, Data Science, Computational Neuroscience

Indian Institute of Science Education & Research (IISER) Tirupati, India
BS-MS Dual Degree (Biology & Physics) **August 2021**
CGPA: 8.2/10.0
Relevant Coursework: Nonlinear Dynamics & Chaos, Linear Algebra (I & II), Calculus (Univariate and Multivariate), Statistical Thermodynamics, Data Science

Specialized Training
Radical AI Founders Masterclass, Radical Ventures (Virtual) **2026**
Methods in Computational Neuroscience (MCN), Marine Biological Laboratory, Woods Hole, MA **2026**

RESEARCH EXPERIENCE

Clayspace Lab, New York University **September 2021 – May 2026**
Doctoral Researcher, Supervisor: Dr. Clayton Curtis

- High-Performance Computing: Built a JAX-based GPU toolbox for population receptive field (pRF) modeling of fMRI data (compressive spatial summation model) that fits a full volumetric dataset in about 5 minutes on a single GPU.
- Multimodal Data Fusion: Led end-to-end pipelines for high-dimensional datasets spanning human behavior, eye-tracking and neural signals (fMRI, MEG, ECoG, EEG).
- Causal Neuromodulation: Ran electric-field simulations (SimNIBS) and TMS experiments to test the causal role of distributed representations in working memory.

- Publications & Peer Review: First author of *Perturbing human V1 degrades the fidelity of visual working memory* (Nature Communications, 2025) and *Neural synchrony between prefrontal and visual cortex supports visual working memory* (bioRxiv, 2026); ad-hoc reviewer for eLife, ICLR, Brain Structure & Function, CogSci and CCN.
- Scientific Communication: Presented at Stanford, Dartmouth, SfN, CCN and the Working Memory Symposium; won NYU's Dean's Award for Research Communication Excellence.

Sciurid Lab, IISER Tirupati **August 2020 – August 2021**
Master's Thesis, Supervisor: Dr. Nandini Rajamani

- Acoustic Deep Learning: Developed CNN and template-matching classifiers to detect rare species' vocalizations in noisy rainforest recordings.
- Citizen Science & Big Data: Managed a national-scale project to ingest, annotate and analyze large acoustic datasets in Python and R.
- Statistical Modeling: Ran hypothesis testing and feature extraction on passive acoustic data for the Master's thesis.

iGEM Team Lead, IISER Tirupati **January 2018 – November 2019**

- Technical Leadership: Led IISER Tirupati's first iGEM team of 15 undergraduates to engineer *CoCa coli*, a probiotic immunotherapy for colon cancer, earning a Gold Medal at the Giant Jamboree in Boston (300+ teams).
- Funding & Execution: Secured INR 1 million (∼$12,000) through the Indian Biological Engineering Competition (iBEC) and ran genetic-circuit design through in-vitro validation.

PROJECTS

- **Proxenos** (mindemory.io/proxenos): Built an open-source, privacy-first AI agent that keeps a person's records on their own computer and drafts emails, forms and follow-ups for them to send, with a desktop app for Mac and Windows; Python, Next.js, local models and the Claude API.
- **Thanos** (mindemory.io/thanos): Building a lab's searchable memory, from grant and account details to equipment and where the code lives, so what a group knows outlasts turnover; started for the Murray Lab (Dartmouth) and the Clay Space Lab (NYU); Next.js, TypeScript, Postgres.
- **MCN Network Graph** (mindemory.io/projects/mcn-network): Built an interactive 2D and 3D network (D3.js, three.js) of every Methods in Computational Neuroscience cohort at MBL, linking people by co-attendance and co-authorship (OpenAlex).
- **Notion Paperpile Integration** (github.com/mindemory/notion_paperpile): Open-source tool that turns a Paperpile library into a Notion reading list and maps it by topic and co-authorship; Python, Notion API, NetworkX.
- **Alexαndria** (alexandriasocial.net): Prototyped a platform that joins shared bibliographies with a collaborative activity space for research teams.

VOLUNTEER WORK

Website Design Coordinator, ClimateMatch Academy **August 2023 – December 2023**

- Cross-Institutional Collaboration: Designed and maintained the website for the ClimateMatch curriculum with Neuromatch Academy, streamlining content delivery for participants worldwide.

Cognition & Perception Open House Student Organizer, New York University **September 2022 – June 2023**

- Event Leadership: Elected to run recruitment for the incoming 2023 PhD cohort: travel, housing and interview scheduling for 40+ international candidates.

AWARDS

Society for Neuroscience Trainee Professional Development Award ($1,000)	**2025**
Dean's Award for Research Communication Excellence, NYU ($2,000)	**2025**
GSAS Conference Travel Grant, NYU	**2025**
MacCracken Doctoral Fellowship, NYU	**2021 – 2026**
Indian Biological Engineering Competition (iBEC) Grant (∼$12,000 each)	**2019, 2020**
INSPIRE Fellowship, Department of Science and Technology India	**2016 – 2021**
Regional Winner, 5th Place, Indian National Brain Bee (INBB)	**2015**

SKILLS

Programming
Python, MATLAB, R, SQL, LaTeX, CSS, JavaScript, HTML, React, NodeJS, Supabase, Vercel

Neuroscience Tools
EEG, MEG, fMRI, TMS, ECoG, Fieldtrip, MNE-Python, SimNIBS, Psychtoolbox

Data Science & ML
JAX, PyTorch, TensorFlow, Keras, Scikit-learn, Presto, Spark, Neural Networks, Time-series Analysis

Other Tools & Languages
Git/GitHub, Adobe Illustrator, Adobe Photoshop
English (Fluent), Hindi, Marathi